Magic Square of Squares

1- Magic Square

We call 3X3 square arrangement of natural numbers a magic square if

1- **NO repeated natural numbers**
2- **The sum of number in each Row = the sum of numbers in each column**
3- **The sum of numbers in both diagonals = sum of numbers in each column and each row**

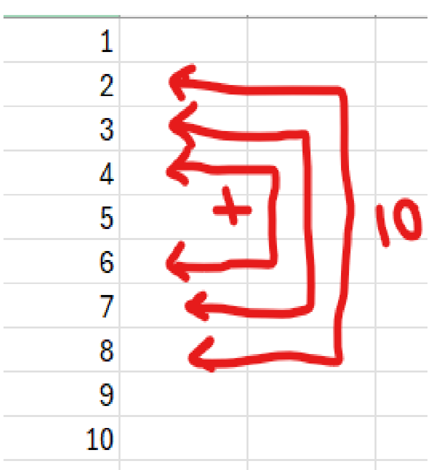
1									
2									
3									
4									
5									
6									
7									
8					33	33	33		
9									33
10					14	10	9		33
11					6	11	16		33
12					13	12	8		33
13									33
14									
15									
16									
17									
18									
19									
20									

Figure 1: Magic Square 3X3 based on natural number 11

11									
12									
13									
14									
15									
16					57	57	57		
17									57
18					22	18	17		57
19					14	19	24		57
20					21	20	16		57
21									57
22									
23									
24									
25									
26									
27									
28									
29									
30									

Figure 2: Magic Square 3X3 based on natural number 19

As it is shown in both figure 1 and figure 2 the numbers in each row and column and both diagonals are on both sides of the base number and with equal spaces because the numbers are separated by difference equal one.



2- Magic Square of Squares

the main difference between the square of squares and the normal magic square is the domain of the numbers used to fill the square

Any square natural number end with {0 or 1 or 4 or 5 or 6 or 9} in the ones place which are only 5 different digits excluding the zero (each one of them can be a result of multiple squares.

3X3 SQAURE must meet the same conditions of the magic square

1- NO repeated natural numbers.
2- The sum of number in each Row = the sum of numbers in each column.
4- The sum of numbers in both diagonals = sum of numbers in each column and each row.
5- All numbers are squares

we need only 5 variables {X1, X2, X3, X4, X5} each are a square value

We will start with one horizontal center row and one right diagonal (Yellow values)

X1+X3 = X4 +X5
X2 =1/2 *(X4+X5)
X2 =1/2 *(X1+X3)

Diag2				Diag 1
	=X1+X2-X4	=X3+X4-X1	X1	
Horizontal1 --->	X4	X2	X5	
Horizontal2--->	X3	=2*X1-X4	=X2+X3-X5	

Diag2 = Diag1	X1+3 * X2+X3-X4-X5 = X1+X2+X3
Diag2 = Horizontal1	X1+3 * X2+X3-X4-X5 = X4+X2+X5

Horizontal 2 = Diag1	X3+2 * X1-X4+X2+X3-X5 = X1+X2+X3
Horizontal 2 =Horizontal 1	X3+2 * X1-X4+X2+X3-X5 = X4+X2+X5

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STEP (1) X1+X2+X3 = X4+X2+X5

STEP (2) X2 is shared between the horizontal row and diagonal and it is added in the SUM of both, so X2 is the average of its row and diagonal values i.e.

$X2 = 0.5 * (X4+X5)$ and $X2 = 0.5 * (X3 + X1)$

STEP (3) We know X1+X2+X3 = SUM of the magic SQUARE

Columns must add up to the same SUM value then
X3 + X4 + Missig value in Column one = SUM
Then Missing value in Column one = X1+X2-X4

STEP (4) Same will be for the third Column, we know X1+X2+X3 = SUM of the magic SQUARE. Columns must add up to the same SUM value then

X1 + X5 + Missig value in Column three = SUM
Then Missing value in Column three = X2+X3-X5

STEP (5) after step 3 we will be able to calculate the final missing value in middle cell in row one

X1+X2-X4 + missing value in row one +X1= SUM

Then Missing value in Row one = X3+X4-X1

SETP (6) after step 4 we will be able to calculate the final missing value in middle cell in row three

X3+ missing value in row three +X2+X3-X5 = SUM

Then Missing value in Row Three = 2 * X1- X4

These calculations guarantee equal sums for all columns and rows and both diagonals and only X1, X2, X3, X4, X4 are squares. And all are different numbers

But breaks the fifth condition not all values are squares

This first arrangement using (X1, X2, X3 , X4 ,X5) lead to three condition equations

$X4 + X3 = X1+X5 \rightarrow (1)$

$2* X2 = X4 +X5 \rightarrow (2)$ X2 is the average of (X4, X5)

$2*X2=X1+X3\rightarrow(3)$ X2 is the average of (X1, X3)

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We will reverse the arrangement for the squares to select the opposite diagonal and center column instead of center row so here (X3, X4, X5, X1) are the missing values in this arrangement.

Diag2				Diag 1
	X6	X8	X1	
Horizontal1 --->	X4	X2	X5	
Horizontal2--->	X3	X9	X7	

Values from First arranegment

X9 = 2*X1-X4

X6=X1+X2-X4

X8=X3+X4-X1

X7=X2+X3-X5

In this arranegment

SUM = X6+X2+X7

X3 = SUM -X9-X7

X3=X6+X2+X7-X9-X7

X3=X6+X2-X9

But from first arragment --> X6=X1+X2-X4

X3=X1+X2-X4+X2-X9

X3=X1+ 2* X2-X4-X9

'But from first arragment --> 'X9 = 2*X1-X4

X3=X1+ 2* X2-X4-(2*X1-X4)

X3=X1 contradiction

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And as it is shown here this means we will have a contradiction for one of the magic square of squares (repeated values) X3 = X1 in order to get the sum of the rows and the columns and both diagonals be equal.

Rearrange EQ (2) X4 = 2*X2 -X5 →(4)

Substitute EQ (4) in EQ (1)

2*X2 -X5 +X3 = X1 +X5

2*X2+X3 -X1 = 2*X5

2*X2 = X1+X3

X1+X3 +X3 -X1 = 2*X5

2* X3 = 2*X5

X3 = X5 and this means we will have repeated values in the magic square, and this is not allowed

And this proof we can not have a magic Square of squares.

3- <u>Patterns in Square Natural Numbers</u>

Any square natural number ends with {0 or 1 or 4 or 5 or 6 or 9} in the ones place, then these are the numbers that will have the most significance in finding the sum of rows and columns and both diagonals.

Note that numbers like {100, 120 ,2000, 5000, ...} all ends by 0

And numbers like {14 ,64 ,144, ...} all ends by 4

Therefore, each digit in this list {0,1,4,5,6,9} represents multiple of other squares.

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X	X^2	1
1	1	1
9	81	1
11	121	1
19	361	1
21	441	1
29	841	1
31	961	1
39	1521	1
41	1681	1
49	2401	1
51	2601	1
59	3481	1
61	3721	1
69	4761	1
71	5041	1
79	6241	1
81	6561	1
89	7921	1
91	8281	1
99	9801	1
101	10201	1
109	11881	1
111	12321	1
119	14161	1
121	14641	1
129	16641	1
131	17161	1
139	19321	1
141	19881	1
149	22201	1
151	22801	1

X	X^2	4
2	4	4
8	64	4
12	144	4
18	324	4
22	484	4
28	784	4
32	1024	4
38	1444	4
42	1764	4
48	2304	4
52	2704	4
58	3364	4
62	3844	4
68	4624	4
72	5184	4
78	6084	4
82	6724	4
88	7744	4
92	8464	4
98	9604	4
102	10404	4
108	11664	4
112	12544	4
118	13924	4
122	14884	4
128	16384	4
132	17424	4
138	19044	4
142	20164	4
148	21904	4
152	23104	4

X	X^2	5
5	25	5
15	225	5
25	625	5
35	1225	5
45	2025	5
55	3025	5
65	4225	5
75	5625	5
85	7225	5
95	9025	5
105	11025	5
115	13225	5
125	15625	5
135	18225	5
145	21025	5
155	24025	5
165	27225	5
175	30625	5
185	34225	5
195	38025	5
205	42025	5
215	46225	5
225	50625	5
235	55225	5
245	60025	5
255	65025	5
265	70225	5
275	75625	5
285	81225	5
295	87025	5
305	93025	5

X	X^2	6
4	16	6
6	36	6
14	196	6
16	256	6
24	576	6
26	676	6
34	1156	6
36	1296	6
44	1936	6
46	2116	6
54	2916	6
56	3136	6
64	4096	6
66	4356	6
74	5476	6
76	5776	6
84	7056	6
86	7396	6
94	8836	6
96	9216	6
104	10816	6
106	11236	6
114	12996	6
116	13456	6
124	15376	6
126	15876	6
134	17956	6
136	18496	6
144	20736	6
146	21316	6
154	23716	6

X	X^2	
3	9	9
7	49	9
13	169	9
17	289	9
23	529	9
27	729	9
33	1089	9
37	1369	9
43	1849	9
47	2209	9
53	2809	9
57	3249	9
63	3969	9
67	4489	9
73	5329	9
77	5929	9
83	6889	9
87	7569	9
93	8649	9
97	9409	9
103	10609	9
107	11449	9
113	12769	9
117	13689	9
123	15129	9
127	16129	9
133	17689	9
137	18769	9
143	20449	9
147	21609	9
153	23409	9

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	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
2	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50	52	54	56	58	60
3	3	6	9	12	15	18	21	24	27	30	33	36	39	42	45	48	51	54	57	60	63	66	69	72	75	78	81	84	87	90
4	4	8	12	16	20	24	28	32	36	40	44	48	52	56	60	64	68	72	76	80	84	88	92	96	100	104	108	112	116	120
5	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	100	105	110	115	120	125	130	135	140	145	150
6	6	12	18	24	30	36	42	48	54	60	66	72	78	84	90	96	102	108	114	120	126	132	138	144	150	156	162	168	174	180
7	7	14	21	28	35	42	49	56	63	70	77	84	91	98	105	112	119	126	133	140	147	154	161	168	175	182	189	196	203	210
8	8	16	24	32	40	48	56	64	72	80	88	96	104	112	120	128	136	144	152	160	168	176	184	192	200	208	216	224	232	240
9	9	18	27	36	45	54	63	72	81	90	99	108	117	126	135	144	153	162	171	180	189	198	207	216	225	234	243	252	261	270
10	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150	160	170	180	190	200	210	220	230	240	250	260	270	280	290	300
11	11	22	33	44	55	66	77	88	99	110	121	132	143	154	165	176	187	198	209	220	231	242	253	264	275	286	297	308	319	330
12	12	24	36	48	60	72	84	96	108	120	132	144	156	168	180	192	204	216	228	240	252	264	276	288	300	312	324	336	348	360
13	13	26	39	52	65	78	91	104	117	130	143	156	169	182	195	208	221	234	247	260	273	286	299	312	325	338	351	364	377	390
14	14	28	42	56	70	84	98	112	126	140	154	168	182	196	210	224	238	252	266	280	294	308	322	336	350	364	378	392	406	420
15	15	30	45	60	75	90	105	120	135	150	165	180	195	210	225	240	255	270	285	300	315	330	345	360	375	390	405	420	435	450
16	16	32	48	64	80	96	112	128	144	160	176	192	208	224	240	256	272	288	304	320	336	352	368	384	400	416	432	448	464	480
17	17	34	51	68	85	102	119	136	153	170	187	204	221	238	255	272	289	306	323	340	357	374	391	408	425	442	459	476	493	510
18	18	36	54	72	90	108	126	144	162	180	198	216	234	252	270	288	306	324	342	360	378	396	414	432	450	468	486	504	522	540
19	19	38	57	76	95	114	133	152	171	190	209	228	247	266	285	304	323	342	361	380	399	418	437	456	475	494	513	532	551	570
20	20	40	60	80	100	120	140	160	180	200	220	240	260	280	300	320	340	360	380	400	420	440	460	480	500	520	540	560	580	600
21	21	42	63	84	105	126	147	168	189	210	231	252	273	294	315	336	357	378	399	420	441	462	483	504	525	546	567	588	609	630
22	22	44	66	88	110	132	154	176	198	220	242	264	286	308	330	352	374	396	418	440	462	484	506	528	550	572	594	616	638	660
23	23	46	69	92	115	138	161	184	207	230	253	276	299	322	345	368	391	414	437	460	483	506	529	552	575	598	621	644	667	690
24	24	48	72	96	120	144	168	192	216	240	264	288	312	336	360	384	408	432	456	480	504	528	552	576	600	624	648	672	696	720
25	25	50	75	100	125	150	175	200	225	250	275	300	325	350	375	400	425	450	475	500	525	550	575	600	625	650	675	700	725	750
26	26	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520	546	572	598	624	650	676	702	728	754	780
27	27	54	81	108	135	162	189	216	243	270	297	324	351	378	405	432	459	486	513	540	567	594	621	648	675	702	729	756	783	810
28	28	56	84	112	140	168	196	224	252	280	308	336	364	392	420	448	476	504	532	560	588	616	644	672	700	728	756	784	812	840
29	29	58	87	116	145	174	203	232	261	290	319	348	377	406	435	464	493	522	551	580	609	638	667	696	725	754	783	812	841	870
30	30	60	90	120	150	180	210	240	270	300	330	360	390	420	450	480	510	540	570	600	630	660	690	720	750	780	810	840	870	900
31	31	62	93	124	155	186	217	248	279	310	341	372	403	434	465	496	527	558	589	620	651	682	713	744	775	806	837	868	899	930

1- Any natural number ends with digit 2 or 8 its square will end by digit 4

2- Any natural number ends with digit 3 or 7 its square will end by digit 9

3- Any natural number ends with digit 1 or 9 its square will end by digit 1

4- Any natural number ends with digit 0 its square will end by digit 00

5- Any natural number ends with digit 5 its square will end by digit 25

If we checked the 3X3 square arrangement for only the ones places for each square number
We will find many arrangements that must have at least one row or one column or one diagonal that do not add up to the Square SUM
except the square that have squares that ends with (5,9,1) in the one place or squares of squares that all values end with the same value at one's place digit these are some of these arrangements

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15	15	15		
				15
5	9	1		15
1	5	9		15
9	1	5		15
				15

X2	Y2	Z2		15	
5	5	5		15	0
5	5	5		15	0
5	5	5		15	0
				15	0
15	15	15			
0	0	0			

X2	Y2	Z2		27	
9	9	9		27	0
9	9	9		27	0
9	9	9		27	0
				27	0
27	27	27			
0	0	0			

5	6	9
9	5	6
6	9	5

4	6	9
6	9	4
9	4	6

0	1	5
5	0	1
1	5	0

0	1	9
9	0	1
1	9	0

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0	1	4
4	0	1
1	4	0

5	9	1
9	1	5
1	5	9

0	6	9
9	0	6
6	9	0

4	5	6
6	4	5
5	6	4

9	1	4
1	5	9
4	9	1

5	1	9
9	5	1
1	9	5

AVG   4 . AVG

				586092	
349924	33124	386884		769932	-183840
232324	195364	158404		586092	0
3844	357604	40804		402252	183840
				586092	0
586092	586092	586092			
0	0	0			
591.5437	182	622			
482	442	398			
62	598	202			

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